

Fourier Analysis — Easy Line-by-Line Solution

Your 16-value example explained step by step

1. What is Fourier Analysis?

Fourier Analysis finds repeating patterns (frequencies) inside a signal. For a finite list of observations, we use the **Discrete Fourier Transform (DFT)**.

Easy idea: Signal → Fourier calculation → repeating-pattern information.

2. Given Data

Time	Signal
1	23.0000
2	27.1213
3	20.0000
4	12.8787
5	17.0000
6	22.8787
7	20.0000
8	17.1213
9	23.0000
10	27.1213
11	20.0000
12	12.8787
13	17.0000
14	22.8787
15	20.0000
16	17.1213

There are 16 signal values, so $N = 16$.

3. Where does 320 come from?

Add all 16 signal values.

First 8:

$$23 + 27.1213 + 20 + 12.8787 + 17 + 22.8787 + 20 + 17.1213 = \mathbf{160}$$

Second 8 are identical = **160**.

$$\mathbf{160 + 160 = 320}$$

Therefore the first Fourier coefficient is $X_0 = 320$. The average signal is $320 \div 16 = \mathbf{20}$.

4. The calculation behind 0 and 23.99988

For each frequency k , Fourier multiplies the signal values by cosine and sine patterns, then adds the products.

$$\text{DFT formula: } X_k = \sum [x_n (\cos(2\pi kn/N) - i \sin(2\pi kn/N))]$$

Do not calculate the whole formula at once. For the real part, use: **Signal $\times$ cosine**, then add.

5. Where does the 0 come from?

At some frequencies, positive and negative products cancel each other. For $k = 1$, the real and imaginary totals are essentially zero, so:

$$X_k \approx 0$$

6. Where does 23.99988 come from?

For $k = 2$ and $N = 16$:

$$2\pi kn/N = 2\pi(2)n/16 = \pi n/4$$

The cosine values are approximately: 1, 0.70710678, 0, -0.70710678, -1, -0.70710678, 0, 0.70710678, and then the same pattern repeats.

n	Signal	$\cos(\pi n/4)$	Signal $\times$ cos
0	23.0000	1.00000000	23.00000000
1	27.1213	0.70710678	19.17765514
2	20.0000	0.00000000	0.00000000
3	12.8787	-0.70710678	-9.10661610
4	17.0000	-1.00000000	-17.00000000
5	22.8787	-0.70710678	-16.17768391
6	20.0000	-0.00000000	-0.00000000
7	17.1213	0.70710678	12.10658733
8	23.0000	1.00000000	23.00000000
9	27.1213	0.70710678	19.17765514
10	20.0000	0.00000000	0.00000000
11	12.8787	-0.70710678	-9.10661610
12	17.0000	-1.00000000	-17.00000000
13	22.8787	-0.70710678	-16.17768391
14	20.0000	-0.00000000	-0.00000000
15	17.1213	0.70710678	12.10658733

Adding the last column gives approximately **23.999884919448**, which is about **24**. Excel may show **23.9998849194482** because the input values were rounded.

7. Why do you see $-40i$ and $+40i$?

Fourier also uses the sine part. The sine total becomes the imaginary part. For your data, $k = 4$ gives approximately:

$$X_{\text{sin}} \approx -40i$$

The matching coefficient at $k = 12$ is approximately:

$$X_{\text{sin}} \approx +40i$$

Here $i = \sqrt{-1}$. Complex numbers are normal in Fourier analysis; they store amplitude and phase information.

8. Complete Fourier Output

k	Real	Imaginary	Coefficient
0	320.000000	0.000000	320.000000
1	0.000000	0.000000	0.000000
2	23.999885	0.000000	23.999885
3	0.000000	0.000000	0.000000
4	0.000000	-40.000000	-40.000000i
5	0.000000	0.000000	0.000000
6	0.000000	0.000000	0.000000
7	0.000000	0.000000	0.000000
8	0.000000	0.000000	0.000000
9	0.000000	0.000000	0.000000
10	0.000000	0.000000	0.000000
11	0.000000	0.000000	0.000000
12	0.000000	40.000000	40.000000i
13	0.000000	0.000000	0.000000
14	23.999885	0.000000	23.999885
15	0.000000	0.000000	0.000000

9. Easy Method to Remember

For each k: (1) take the signal, (2) multiply by the cosine pattern, (3) multiply by the sine pattern, (4) add the products, (5) combine real and imaginary parts.

Remember: 320 comes from adding all 16 values. The 0 and 23.99988 come from multiplying the signal by Fourier's special cosine/sine patterns and then adding the results.